Diagnose, repair, recheck

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify a reasoning error and demonstrate the repaired method on a new problem.

SUCCESS CRITERIA - I can...

1. Locate the first incorrect step.
2. Show a corrected method with units.
3. Apply the repair to a different problem.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “evidence” and “revision” to explain your model or result. Sentence starters are optional.

evidence: observations or measurements that support an idea

revision: a change made after checking evidence or reasoning

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A 3 kg cart moves left at 5 m/s. Forces are 26 N right and 5 N left.
Find acceleration and whether its speed initially rises or falls.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$.
For a constant-acceleration interval. Check signs, units and assumptions; keep the original step visible when revising.

Worked reasoning: Claim: a car moving east must accelerate east. Counterexample: it moves east while braking, so acceleration points west. A single valid counterexample can disprove a universal claim.

Your turn: Draw velocity and acceleration arrows for that counterexample. Write a corrected statement about their directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Car braking while moving east

East is right.



Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③

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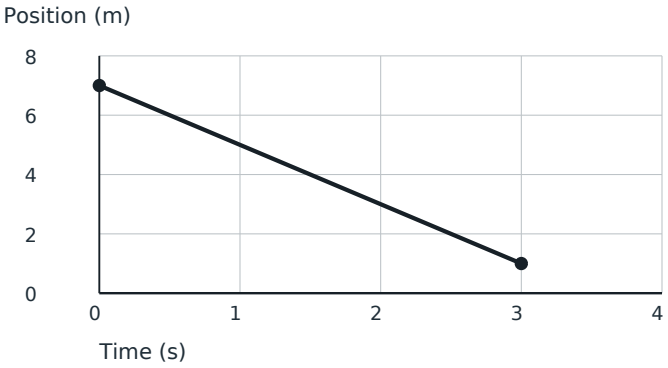
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Position is not velocity

A cart's position changes from 7 m to 1 m in 3 s, as shown. A student says, "Its velocity is 1 m/s because it ends at 1 m." Explain the mistake and calculate the velocity correctly.

Sentence starter: The step I would change is … . The evidence is … .



Marzano check - circle: ① ② ③

04 Net force is not one force

A 4 kg cart has 18 N right and 6 N left. A student calculates acceleration as 18 ÷ 4.
Use the force diagram to explain the missing step, then correct the calculation.

Sentence starter: My claim is … . The diagram/data show …, so … .

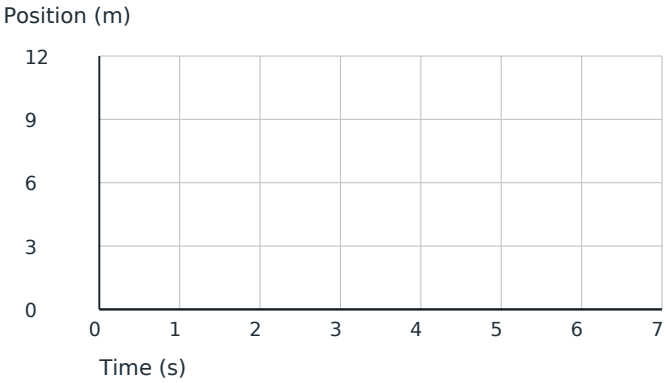


Marzano check - circle: ① ② ③

05 New evidence

Try two new checks.
(a) A robot moves steadily from 10 m to 2 m in 7 s.
Find its velocity.
(b) A 5 kg cart has 7 N right and 22 N left.
Find its acceleration.
Show how each calculation accounts for direction.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

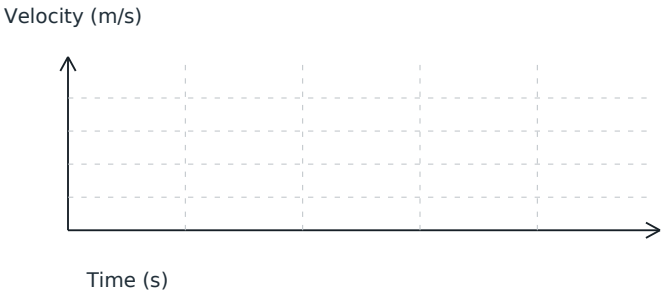
Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A cart travels west at 14 m/s and west at 5 m/s 4 s later. Is it speeding up?
Find acceleration size and direction.
Sketch the velocity-time model; label the given values.

Word help: acceleration = rate of change of velocity

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

07 Exit check

Choose the error in question 3 or 4.
Explain the correction, then point to the matching new check in question 5. What can you now do without help?

Sentence starter: The step I would change is ... . The evidence is ... .

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Invent a wrong statement about force or motion, then draw a specific case that disproves it.
State the corrected rule.

Sentence starter: The step I would change is ... . The evidence is ... .



Add and label force arrows.

Use a diagram to test the statement you invent.

Marzano check - circle: 1 2 3